

Flood Roofing — gable engine: ridge-offset slider + gable-end picker

Eight cases exercising the two new engine fields: `DRAW.gableRidgeOffset` (signed $-1..+1$, slides the ridge along the perpendicular axis) and `DRAW.gableEnds` (array of outline edge indices to keep as rake barges; remainder become hip-ends). Per-cell caption shows the lines emitted + the per-face sheet length the strip engine derived from them. Cases A–D show the ridge slider on a plain rectangle (notice the per-face sheet length splits as soon as $\text{offset} \neq 0$). E–F compare auto vs manual gable-end picking on an L-shape. G–H apply the slider to a rectangle-with-wing outline.

A — Ridge centred (offset 0)

24 sheets · 48 strips

900 × 600 rectangle, ridge at midline — baseline gable.

Lines: ridge · 2×gutter · 2×barge:finish · 2×barge:starter · 4×barge

Faces: long-side: 24 sheets × 6.47m long-side: 24 sheets × 6.47m

B — Ridge slid +0.3 (toward south gutter)

24 sheets · 48 strips

Same 900×600. Ridge offset = +0.3 → north face gets longer sheets, south face shorter.

Lines: ridge · 2×gutter · 2×barge:finish · 2×barge:starter · 4×barge

Faces: long-side: 24 sheets × 8.41m long-side: 24 sheets × 4.53m

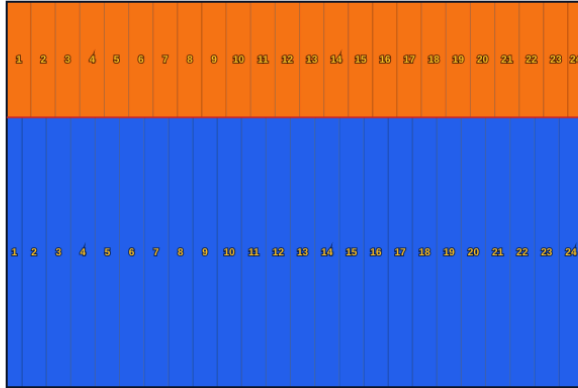
C — Ridge slid -0.4 (toward north gutter)

24 sheets · 48 strips

Same 900×600. Ridge offset = -0.4 → flipped: north shorter, south longer.

Lines: ridge · 2×gutter · 2×barge:finish · 2×barge:starter · 4×barge

Faces: long-side: 24 sheets × 3.88m long-side: 24 sheets × 9.06m



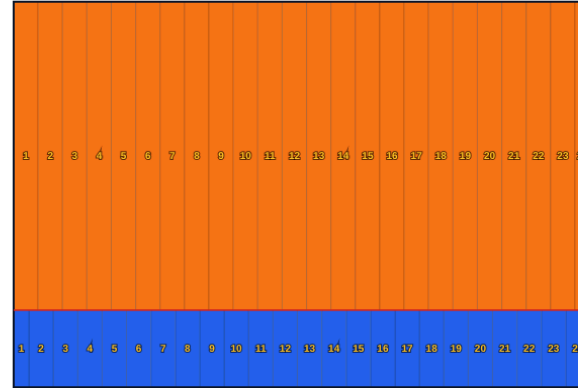
D — Aggressive +0.6 slide

24 sheets · 48 strips

Same 900×600. Ridge offset = +0.6 → very asymmetric gable, one face dominant.

Lines: ridge · 2×gutter · 2×barge:finish · 2×barge:starter · 4×barge

Faces: long-side: 24 sheets × 10.35m long-side: 24 sheets × 2.59m



E — L-shape (canonical), auto ends

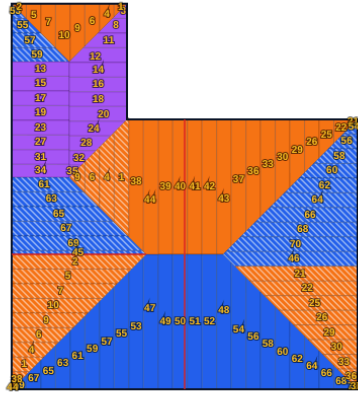
70 sheets · 111 strips

Wing 300×300 / main 900×700. Engine auto-classifies the L into long-side + hip-end faces.

Lines: ridge · 3×gutter · 5×barge · 2×barge:starter · 2×barge:finish

Faces:

long-side: 70 sheets × 7.55m



F — L-shape, gable end on edge 0 only

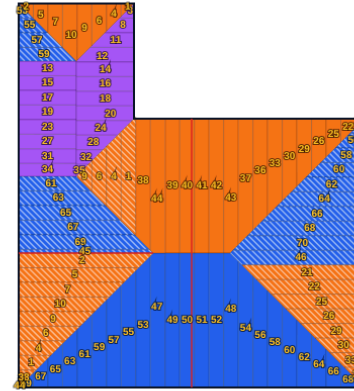
70 sheets · 111 strips

DRAW.gableEnds=[0] — only the wing-top rake becomes a barge; perpendicular edges 1,3,5 fall through to hip-end treatment (no barge stamp).

Lines: ridge · 3×gutter · 5×barge

Faces:

long-side: 70 sheets × 7.55m



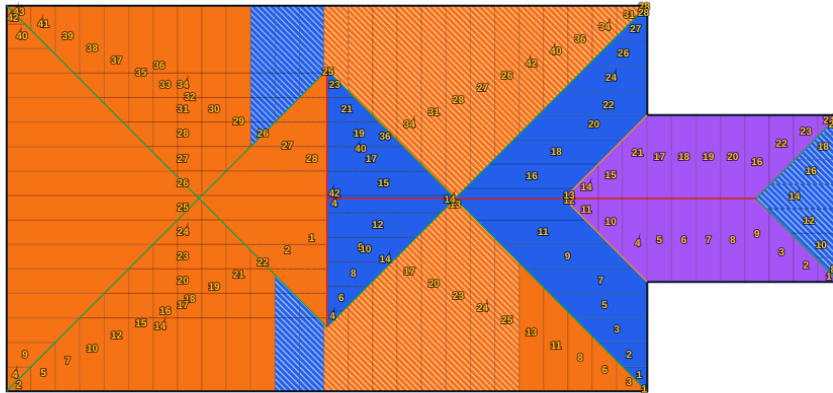
G — Rectangle with east wing, auto

43 sheets · 129 strips

Main 1000×600 + wing 300×260 at y=170. Auto engine picks ridge along long axis.

Lines: ridge · 4×gutter · 6×barge · 2×barge:finish · 2×barge:starter

Faces: hip-end: 43 sheets × 10.79m



H — Rectangle with wing + ridge slid +0.25

43 sheets · 129 strips

Same main+wing outline; ridge slid toward south gutter. Main face sheets become non-symmetric.

Lines: ridge · 4×gutter · 6×barge · 2×barge:finish · 2×barge:starter

Faces: hip-end: 43 sheets × 10.79m

